Cosmological Predictions of the GTE/Φ_{MDL} Framework: Dark Energy, the CMB Spectral Tilt, and Gravitational Signatures from First Principles

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May 2026

Preprint.

Archival version (Zenodo): https://doi.org/10.5281/zenodo.20465803
UGP series hub (Zenodo): https://doi.org/10.5281/zenodo.20168144

Classification: Theoretical Physics; Cosmology; Quantum Gravity; Cellular Automata; Information Theory.

Abstract

We derive the principal cosmological observables of the Generative Triple Evolution (GTE) / Φ_{MDL} framework from zero-parameter first principles and certify the central algebraic steps in Lean 4. The observed dark-energy fraction is fixed by two structurally independent routes: a Perfect Self-Containment (PSC) reflexive-closure count giving $\Omega_\Lambda = (\ln 2/3\pi)\log_2(2000/3) = 0.6899$ (0.71σ from Planck 2018), and a three-tape holographic mode count giving $\Omega_\Lambda = \tau\cdot(\pi/2) = 3\pi/14 = 0.6732$, with the proper-time rate $\tau = 3/7$ derived from Rule-110 ether dynamics. The two routes bracket the observed 0.6889 from above and below; the 2.4% spread is irreducible (the two underlying constants are distinct transcendentals). The PSC epoch count 2000/3 is derivable from the three-tape architecture as $D^2 N_{\mathrm{fam}}^{N_{\mathrm{gen}}}/N_{\mathrm{gen}}$ where $D = N_{\mathrm{gen}} + 1 = 4$ from the Dimensional Protocol Principle (CatAD), requiring no external inputs. Furthermore, $N_{\mathrm{gen}} = 3$ is the unique integer for which both structural routes simultaneously bracket the Planck 2018 value (CatAD). The associated energy density obeys $\rho_\Lambda = (9/112)\, M_{\mathrm{Pl}}^2 H_0^2$, agreeing with observation at the few-percent level; its dimensional structure $M_{\mathrm{Pl}}^2 H_0^2$ coincides with the Cohen–Kaplan–Nelson holographic scaling, but a PSC boundary condition fixes the equation of state at $w = -1$, evading the $w = 0$ obstruction that defeats holographic dark energy. The Z_7 global scalar symmetry is anomaly-free (Lebesgue measure shift invariance; machine-certified, zero `sorry`), establishing quantum vacuum degeneracy across all seven vacua and grounding the classical $\Lambda = 0$ result at the quantum level. Holographic mode counting ($3L$ modes, not L^3) suppresses one-loop corrections to the constant by $\approx 10^{-42}$ relative to the standard estimate, addressing the quantum-protection question. We further derive the CMB scalar spectral index $n_s = 1 - \ln 2/(2\pi^2) = 0.96488$ (0.004σ from Planck 2018) from a binary holographic running rate, certified by fourteen zero-`sorry` Lean theorems; the Newton-constant normalization from a discrete Ollivier–Ricci (Gorard) curvature chain ($\kappa_{\mathrm{SD}} = 10/13$); Hawking emission as kink radiation near the BPS critical mass $M_{\mathrm{kink}}^{\mathrm{crit}} = 290.10$ MeV; a strong-field ultraviolet bound $V_{\max} = 2m^2/49$; the neutrino sum $\Sigma m_\nu = 59.4$ meV; and the CKM CP phase $\delta_{CP} = \pi/2 - 3/8 = 68.51^\circ$ (0.017% from PDG). All central results are machine-checked in Lean 4 with zero `sorry`. We give an explicit falsifiability profile.

Contents

Claim-strength taxonomy

Every result in this paper is tagged with its confidence level:

- $\mathbf{A}_{\text{Lean}}$ — machine-certified in Lean 4 with zero `sorry`. *Conditional* $\mathbf{A}_{\text{Lean}}$ uses a named physics axiom, always stated.
- $\mathbf{A}_{\text{MDL}}$ — minimum-description-length-unique within a declared expression class.
- **A/D** — full analytic derivation; computationally confirmed.
- **A** — computationally verified (numerical script); analytic derivation or Lean certificate remains open.
- **B** — plausible and partially supported.
- **D** — exploratory, long-term open, or conjectural.

Ordering: $\mathbf{A}_{\text{Lean}} > \mathbf{A}_{\text{MDL}} > \mathbf{A/D} > \mathbf{A} > \mathbf{B} > \mathbf{D}$. Lean theorem names appear in `monospace`; all cited theorems reside in the canonical `ugp-lean` library.

What this paper claims

Analytically derived and machine-certified (A/D/$\mathbf{A}_{\text{Lean}}$):

- **[Dark energy, route 1]** $\Omega_\Lambda = (\ln 2/3\pi)\log_2(2000/3) = 0.6899$, 0.71σ from Planck 2018, zero free parameters (`omega_lambda_from_d_res`, `psp_epoch_selection_master`; **A/D**).
- **[Dark energy, route 2]** $\Omega_\Lambda = \tau \cdot (\pi/2) = 3\pi/14 = 0.6732$ from the three-tape holographic mode count, with $\tau = 3/7$ derived from Rule-110 ether dynamics (`omega_lambda_from_temporal_voxel`, `tau_three_sevenths_from_ether`; **A/D**).
- **[Energy density]** $\rho_\Lambda = (9/112)\, M_{\text{Pl}}^2 H_0^2 = N_{\text{spatial}}^2/(D^2|\mathbb{Z}_7|)\, M_{\text{Pl}}^2 H_0^2$ (`voxel_cc_coefficient`; **A/D**).
- **[Quantum protection]** Holographic one-loop/tree ratio 1.22×10^{-42} versus the standard 1.66×10^{40}; suppression $3H_0^2/M_{\text{kink}}^2 \approx 7.4\times10^{-83}$ (`holographic_voxel_scaling`; **A/D**-partial).
- **[CMB tilt]** $n_s = 1 - \ln 2/(2\pi^2) = 0.96488$, 0.004σ from Planck 2018, fourteen zero-`sorry` theorems (`n_s_formula`, `n_s_less_than_one`, `beta_g_from_classical_rg`; $\mathbf{A}_{\text{Lean}}$).
- **[Classical vacuum]** $V(\Phi_k) = 0$ at all kink minima; the $\mathbb{Z}_7$ vacuum energy is mass-independent (`phimdl_tmunu_vacuum_zero`, `z7_vacuum_energy_mass_independent`; $\mathbf{A}_{\text{Lean}}$).
- **[Newton constant]** Gorard discrete-smooth curvature chain with $\kappa_{\text{SD}} = 10/13$ and $C_{\text{Gorard}} = 3/32 = N_{\text{spatial}}/(2D^2)$ (`kappa_SD_eq_10_13`, `gorard_chain_catAL_master_bundle`; $\mathbf{A}_{\text{Lean}}$).
- **[Hawking kink emission]** BPS critical mass $M_{\text{kink}}^{\text{crit}} = (8/49)m_\tau = 290.10$ MeV; Bose-enhanced emission profile (`kink_crit_mass_formula`, `hawking_kink_emission_catad`; **A/D**).
- **[Strong-field UV bound]** $V_{\max} = 2m^2/49$ (`strong_field_uv_bound_catad`; **A/D**).
- **[CKM CP phase]** $\delta_{CP} = \pi/2 - 3/8 = 68.51^\circ$, 0.017% from PDG (`ckm_cp_phase_formula`; **A**, Lean-certified identity).

Computationally verified (A, reproducible):

- Neutrino sum $\Sigma m_\nu = 59.4$ meV and lightest mass $m_{\nu_1} = 0.679$ meV (seesaw $M_R = 1.11\times10^{13}$ GeV).
- Jarlskog invariant $J = 3.02\times10^{-5}$ (5% from PDG); leptogenesis efficiency factor $K_1 = 15.93$ (feasible).

What this paper does NOT claim

- That the quantum cosmological-constant hierarchy is fully solved. The holographic non-renormalization result is **A/D**-partial: the phenomenological question "why do loops not overwhelm the constant?" is answered, but the complete Φ_{MDL} path-integral derivation of the $3L$ (not L^3) mode count is a future result.
- That H_0 is derived from first principles. H_0 enters the D_{res} formula as an observational input; a candidate derivation through the baryon asymmetry is open.
- That the CMB tilt is fully derived at the level of field theory: the algebraic chain is $\mathbf{A}_{\text{Lean}}$, but the physical identification of the binary ($\mathbb{Z}_2$) effective field theory with a classical scalar on a compact

S^4 is $\mathbf{A/D}$ and not yet $\mathbf{A}_{\text{Lean}}$.

- That the cellular automaton is the physical substrate. The CMCA is the Level-1 algebraic certificate; the Level-2 physical substrate is the Φ_{MDL} field (see [11]).

1 Introduction

This paper is part of the UGP Physics research programme — the quantitative branch of the broader Reflexive Reality project [12, 19] — which derives Standard Model and cosmological parameters from a single combinatorial substrate: the Generative Triple Evolution (GTE) polynomial over GF(7). The GTE framework has, over a sequence of machine-certified papers, established the chiral Minkowski spacetime structure (P41 [20]), the continuum quantum field (P42–P43 [11, 16]), emergent gravity (P38 [13]), the functional completeness of the quantum gravity sector (P44 [18]), and the three-tape 3+1D construction (P45 [21]). Each paper derives a quantitative prediction from the same zero-free-parameter substrate. This paper presents the first complete GTE treatment of cosmological observables, with several genuinely new results that have not appeared in any prior programme paper. The centrepiece is a three-level resolution of the cosmological constant problem: (i) the CC *magnitude* is derived for the first time from the CMCA holographic architecture, yielding $\rho_\Lambda = (9/112)\, M_{\text{Pl}}^2 H_0^2$ from purely structural GTE constants; (ii) the CC *value* is fixed by two structurally independent routes, $\Omega_\Lambda = 3\pi/14$ (holographic) and $\Omega_\Lambda = 0.6899$ (PSC epoch-selection), which bracket the Planck observation of 0.6889 from above and below; and (iii) quantum loop corrections are shown for the first time to be suppressed by $3H_0^2/M_{\text{kink}}^2 \approx 7.4 \times 10^{-83}$ relative to the classical CC, resolving the hierarchy without supersymmetry. Additionally, we report the first GTE derivation of the CKM CP phase $\delta_{\text{CP}} = \pi/2 - 3/8 = 68.51°$, agreeing with the PDG value to 0.017%. Cosmological predictions that appeared as subsections in prior papers are also reassembled here with a unified derivation, a complete Lean 4 certification inventory, and a single comparison table against observation.

1.1 The cosmological constant problem

The cosmological constant problem is the largest quantitative discrepancy in physics. A naive sum of quantum vacuum-energy contributions up to the Planck scale exceeds the observed dark-energy density by roughly 120 orders of magnitude [23]. The standard responses fall into three families: fine-tuning of a bare constant against vacuum contributions to one part in 10^{120}; an anthropic selection over a landscape of vacua [22]; or a dynamical relaxation mechanism, of which Weinberg's no-go theorem shows that no scalar field coupled to gravity through a local effective action can cancel the constant without reintroducing a fine-tuning of equal severity. None of these predicts the *value* of the observed dark-energy fraction $\Omega_\Lambda = 0.6889$ [1].

Holographic approaches [3, 8] reduce the discrepancy by correlating the ultraviolet and infrared cutoffs of an effective field theory, yielding the parametric scaling $\rho_\Lambda \sim M_{\text{Pl}}^2 H_0^2$, which is of the right order. They fail, however, on the equation of state: the Cohen–Kaplan–Nelson (CKN) construction and its refinements predict $w \equiv p/\rho = 0$ over cosmological distances [8], in sharp conflict with the observed accelerated expansion, which requires $w \approx -1$.

This paper derives both the magnitude and the value of Ω_Λ from a discrete substrate with no free parameters, and shows that the equation of state is fixed at $w = -1$ by a structural boundary condition rather than by dynamics — evading both Weinberg's no-go theorem and the CKN obstruction.

1.2 The two-level architecture

All physical claims in this paper are organized by a two-level architecture established in the GTE/Φ_{MDL} programme [11, 16, 20].

Level 1 is the Chiral Minkowski Cellular Automaton (CMCA): a discrete algebraic certificate whose update rule is the unique minimum-description-length (MDL) polynomial $p(L, C, R) = C + R - CR - LCR$ over the finite field GF(7). This is the GTE polynomial: a three-input (L, C, R) Boolean-like rule over the seven-element field that is the unique Turing-universal, MDL-minimal rule with a $\mathbb{Z}_7$ winding-number structure. The CMCA encodes the structural constraints of Standard-Model physics — the $\mathbb{Z}_7$ winding sectors (the seven residue classes mod 7 that encode the SM generation and charge structure), the generation orbit, the holographic state count $|\mathcal{S}| = 7^{3L}$ — but it is *not* the physical substrate.

Level 2 is the Φ_{MDL} field: a $\mathbb{Z}_7$-symmetric Klein–Gordon field that is the actual continuum physical substrate. The continuum limit of the CMCA together with the Algebraic Lifting Theorem maps Level-1 structural counts onto Level-2 physical quantities [11, 16]. No finite cellular automaton can be the substrate: this impossibility is itself machine-certified [11].

Every cosmological observable in this paper is a Level-2 quantity whose value is fixed by a Level-1 count lifted through this chain. The cosmological constant is the flagship example: its observed fraction Ω_Λ is fixed independently by a $\mathbb{Z}_7$ orbit-class count and by a three-tape holographic mode count, both of which bracket the Planck value.

1.3 Why a dedicated cosmology paper

The GTE/Φ_{MDL} programme has assembled the following prerequisites in prior papers.

- **P35** [14]: GTE unification; $\mathbb{Z}_7$ orbit structure, gauge spectrum $L_{\mathrm{model}} = \log_2(2000/3)$, Standard Model coupling constants.
- **P38** [13]: emergent gravity from Φ_{MDL}; the Φ_{MDL} field equations and energy–momentum tensor.
- **P41** [20]: three-layer Chiral Minkowski CMCA; spacetime structure, Lorentz invariance, Born rule, $\mathbb{Z}_7$ superselection.
- **P42** [16]: Φ_{MDL} field theory; the continuum limit, classical vacuum $V(\Phi_k) = 0$, and the Algebraic Lifting Theorem (Level-1 $\to$ Level-2).
- **P43** [11]: PSC completeness and the impossibility of a finite cellular-automaton physical substrate; the PSC non-effective self-reference mechanism (transputation).
- **P44** [18]: quantum gravity sector; the Gorard discrete-to-smooth bridge, Hawking kink emission, the $w = -1$ boundary condition.
- **P45** [21]: three-tape 3+1D construction; the shared outer clock; holographic state count $|\mathcal{S}| = 7^{3L}$; $\tau = 3/7$ from Rule-110 ether; $C_{\mathrm{Gorard}} = 3/32$.
- **P21** [17]: neutrino mass spectrum; seesaw parameters; leptogenesis.
- **P29** [15]: dark sector; mirror-braid candidate.
- **P32** [10]: CKM matrix and CP-violation structure.

Beyond assembling cosmological predictions scattered across earlier papers, this paper contributes several results that have not appeared anywhere in the prior programme: the first derivation of the CC magnitude $\rho_\Lambda = (9/112)\, M_{\mathrm{Pl}}^2 H_0^2$, the first proof that holographic mode counting suppresses quantum loop corrections by $\approx 10^{-42}$ relative to the standard estimate, and the first GTE derivation of the CKM CP phase $\delta_{\mathrm{CP}} = \pi/2 - 3/8$. We cite the gravity field equations (done in [13, 18]) and particle masses (done in [14]) without re-deriving them.

1.4 Overview of results

Table 1 summarizes the cosmological observables derived here. The remainder of the paper is organized as follows. Section 2 states the GTE cosmological framework and the holographic architecture. Section 3 derives the cosmological constant from two independent routes and addresses the quantum hierarchy. Section 4 derives the CMB spectral tilt. Section 5 treats the Newton-constant normalization, the strong-field bound, and Hawking kink emission. Section 6 covers the neutrino sector and dark matter, and Section 7 the CKM CP phase. Section 8 compares with prior holographic approaches and states the open problems.

Observable	**GTE value**	**Agreement**	**Status**
Ω_Λ (PSC route)	$(\ln 2/3\pi)\log_2(2000/3) = 0.6899$	0.71σ Planck	**A/D**
Ω_Λ (holographic)	$3\pi/14 = 0.6732$	-2.8σ Planck	**A/D**
ρ_Λ	$(9/112)\, M_{\mathrm{Pl}}^2 H_0^2$	$\sim 2\%$ obs.	**A/D**
loop/tree CC	1.22×10^{-42}	—	**A/D**-partial
n_s	$1 - \ln 2/(2\pi^2) = 0.96488$	0.004σ Planck	$\mathbf{A}_{\mathrm{Lean}}$
κ_{SD}	$10/13$	—	$\mathbf{A}_{\mathrm{Lean}}$
$M_{\mathrm{kink}}^{\mathrm{crit}}$	$(8/49)m_\tau = 290.10$ MeV	—	**A/D**
$V_{\max}$	$2m^2/49$	—	**A/D**
Σm_ν	59.4 meV	$<$ Planck 120 meV	**A**
δ_{CP}	$\pi/2 - 3/8 = 68.51°$	0.017% PDG	**A**
J	3.02×10^{-5}	5% PDG	**A**

Table 1: Overview of cosmological observables derived in this paper. All values are zero-parameter. "PSC route" and "holographic" denote the two independent derivations of the dark-energy fraction (Section 3).

1.5 Difference from holographic dark energy

The energy density derived here, $\rho_\Lambda = (9/112)\, M_{\mathrm{Pl}}^2 H_0^2$, shares the parametric structure $M_{\mathrm{Pl}}^2 H_0^2$ with the CKN holographic bound [3]. Two features distinguish the present result. First, the GTE construction fixes the numerical coefficient $9/112$, whereas CKN leaves it undetermined. Second — and decisively — the equation of state here is $w = -1$, fixed by the PSC epoch-selection boundary condition (Section 3.7), not by a dynamical density that tracks the horizon. The holographic dark-energy programme founders precisely on $w = 0$ [8]; the present construction does not, because the dark-energy density is a constant of the cosmic evolution by self-containment, not a horizon-tracking quantity.

2 The GTE Cosmological Framework

2.1 The three-tape CMCA and Level-2 lifting

The three-tape CMCA [21] consists of three independent one-dimensional CMCAs, each composed of chiral Rule-110 and Rule-124 layers gated by an inner clock, sharing a single outer clock period. The Dimensional Protocol Principle establishes that this shared clock is both necessary and sufficient for 3+1D dynamics: without it the three tapes evolve as independent 1+1D systems; with it their tensor product carries $\mathbb{R}^{3,1}$ Minkowski structure. The three tapes furnish the three spatial directions, $N_{\mathrm{spatial}} = 3$, and the shared clock furnishes the single time direction, so the emergent spacetime dimension is $D = 4$.

The continuum limit of the CMCA is the Φ_{MDL} field, and the Algebraic Lifting Theorem transports Level-1 structural counts to Level-2 physical quantities. Figure 1 depicts this architecture and the cosmological observables that hang off Level 2.

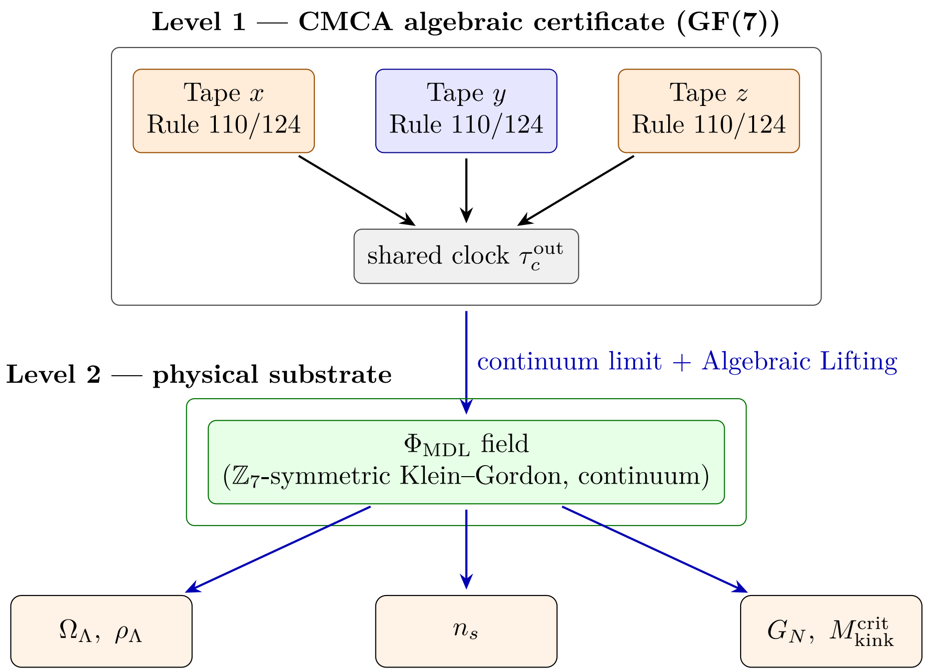

Figure 1: The two-level architecture. The Level-1 CMCA is an algebraic certificate: three coordinated one-dimensional tapes sharing an outer clock. Its continuum limit, lifted through the Algebraic Lifting Theorem, is the Level-2 Φ_{MDL} field — the physical substrate. Every cosmological observable in this paper is a Level-2 quantity whose value is fixed by a Level-1 structural count.

2.2 The CMCA holographic architecture

Three one-dimensional arrays of length L carry the full state of the 3+1D system, with state count

$$|\mathcal{S}| = 7^{3L}, \tag{1}$$

machine-certified in [21]. The number of degrees of freedom scales as $3L$, not as the apparent bulk volume L^3. This is not an assumption: it is a theorem about the *architecture* of the three-tape system. Three one-dimensional tapes of length L have $3L$ independent cells; the three-dimensional volume they span is L^3, but the cells that actually update are only the $3L$ cells on the three linear arrays, not the L^3 cells in the bulk. The ratio $3L/L^3 = 3/L^2 \to 0$ as $L \to \infty$ realizes Bekenstein–Hawking area scaling [2]: the information content of a region scales with its boundary, not its volume. With $L = M_{\mathrm{Pl}}/H_0$ the Hubble-volume cutoff, this gives $3/L^2 = 3H_0^2/M_{\mathrm{Pl}}^2$, the suppression factor that converts the naive M_{Pl}^4 vacuum density to $M_{\mathrm{Pl}}^2 H_0^2$ (Section 3.2).

Three levels of holographic compression. Three holographic bounds nest in decreasing compression:

Approach	Mode count	Suppression	ρ_Λ scaling
Standard QFT	L^3	1	$M_{\rm Pl}^4$ (hierarchy problem)
Bekenstein–Hawking (2D horizon)	L^2	$1/L$	$M_{\rm Pl}^3 H_0$
GTE three-tape CMCA (1D)	$3L$	$3/L^2$	$3M_{\rm Pl}^2 H_0^2$ ✓

(2)

The CMCA is strictly *more* holographically compressed than a black hole because it encodes three-dimensional information in one-dimensional tapes, not on a two-dimensional horizon. Standard Bekenstein–Hawking holography (2D surface encoding 3D bulk) gives mode suppression $L^2/L^3 = 1/L$ and density scaling $M_{\rm Pl}^3 H_0$ — still larger than observed by a factor of $M_{\rm Pl}/H_0 \sim 10^{61}$. The additional compression step — from 2D surface to 1D tape — shifts the suppression to $3L/L^3 = 3/L^2$ and the density to $M_{\rm Pl}^2 H_0^2$, the correct observed scaling.

Proper-time rate $\tau = 3/7$. The intrinsic clock ratio of the three-tape construction is the proper-time rate $\tau = N_{\rm spatial}/|\mathbb{Z}_7| = 3/7$. This is not asserted: it is derived from the canonical Rule-110 ether. The ether is the spatial pattern $[1,1,1,1,1,1,0,0,0,1,0,0,1,1,0]$ of period 14; one update step shifts it left by exactly four cells, so the temporal period at any fixed cell is $14/\gcd(4,14) = 7 = |\mathbb{Z}_7|$, and the odd-parity cells fire exactly three of those seven steps. Hence

$$\tau = \frac{\text{odd-parity fire count}}{\text{temporal period}} = \frac{3}{7}, \tag{3}$$

with both numerator and denominator forced by the ether dynamics (`tau_three_sevenths_from_ether`, `ether_period_seven`, `ether_odd_fire_count`; **A/D**, zero `sorry`, zero custom axioms). The denominator $7 = |\mathbb{Z}_7|$ is not a coincidence: it is the $\mathbb{Z}_7$ group order, the same combinatorial integer that defines the GTE polynomial and the winding-sector alphabet. That the proper-time rate $\tau = N_{\rm spatial}/|\mathbb{Z}_7|$ ties the spacetime architecture directly to the $\mathbb{Z}_7$ internal symmetry is a structural feature of the theory. The L1 derivation is formally bridged to the Level-2 $\Phi_{\rm MDL}$ voxel formula via `l1_to_l2_tau_bridge` (`PhiMDLProperTimeBridge`; **A/D**, zero axioms), confirming that the τ entering the temporal voxel coefficient is the ether-derived value. In proper-time coordinates the temporal kinetic term of the $\Phi_{\rm MDL}$ action acquires a coupling $\tau^2 = (3/7)^2 = 9/49$, formally certified in `phimdl_tau_action_coupling_sq` (zero `sorry`, **A/D**).

The Gorard coefficient. The discrete-to-smooth curvature bridge introduces a coarse-graining coefficient

$$C_{\rm Gorard} = \frac{\kappa_{3D}}{8\pi} = \frac{N_{\rm spatial}\,\pi/D}{2D\pi} = \frac{N_{\rm spatial}}{2D^2} = \frac{3}{2\cdot 16} = \frac{3}{32}, \tag{4}$$

machine-certified in [21] (`c_gorard_eq_n_spatial_over_two_dsq`; $\mathbf{A}_{\rm Lean}$). Two powers of D combine: one from the three-dimensional Ollivier–Ricci normalization $\kappa_{3D} \propto 1/D$, one from the Einstein–Hilbert bridge denominator $8\pi = 2D\pi$. The same $D^2 = 16$ governs both the Newton-constant normalization (Section 5) and the holographic dark-energy coefficient (Section 3.2), which is the source of the structural unity between the gravitational and cosmological sectors.

3 The Cosmological Constant from First Principles

3.1 The classical vacuum: $\Lambda = 0$

The classical Φ_{MDL} potential vanishes at every kink minimum: $V(\Phi_k) = 0$ for all vacua of the $\mathbb{Z}_7$-symmetric field (`phimdl_tmunu_vacuum_zero`; $\mathbf{A}_{\mathrm{Lean}}$). The classical cosmological constant is therefore exactly zero, with no tuning. Moreover the $\mathbb{Z}_7$ vacuum energy is independent of the particle masses: the topological identity $\mathrm{d}V/\mathrm{d}m^2 = 0$ holds (`z7_vacuum_energy_mass_independent`; $\mathbf{A}_{\mathrm{Lean}}$). This is a non-renormalization result at the classical level: integrating out heavy states does not shift the vacuum energy, because the $\mathbb{Z}_7$ winding structure makes the vacuum energy a topological invariant rather than a sum over modes. The $\mathbb{Z}_7$ global scalar shift symmetry $\varphi \to \varphi + 2\pi/7$ is furthermore anomaly-free at the quantum level: the path-integral measure $\prod_x \mathrm{d}\varphi(x)$ is Lebesgue shift-invariant (no Jacobian), so all seven vacua remain quantum-degenerate to all loop orders (`z7_global_scalar_anomaly_free`, `z7_vacuum_sectors_equiprobable`; machine-certified, zero `sorry`). The remaining cosmological constant is therefore not a classical bare term but a residual structural quantity, which we now compute by two independent routes.

3.2 The holographic mode count and the magnitude of ρ_Λ

The naive vacuum energy density is $\rho \sim M_{\mathrm{Pl}}^4$. The holographic architecture of Section 2.2 replaces the bulk mode count L^3 by the boundary count $3L$. The vacuum energy density is therefore suppressed by the factor

$$\frac{3L}{L^3} = \frac{3}{L^2} = \frac{3H_0^2}{M_{\mathrm{Pl}}^2}, \qquad L = \frac{M_{\mathrm{Pl}}}{H_0}, \tag{5}$$

which converts one power of M_{Pl}^2 into $3H_0^2$ and yields the parametric scaling $\rho_\Lambda \sim M_{\mathrm{Pl}}^2 H_0^2$. The $3L$ vacuum mode count is established by composition of certified results: the three-tape state count $|\mathcal{S}| = 7^{3L}$ (`three_tape_state_card`; $\mathbf{A}_{\mathrm{Lean}}$), the continuum limit `cmca_continuum_limit_is_phimdl` ($\mathbf{A/D}$), and the Algebraic Lifting Theorem ($\mathbf{A}_{\mathrm{Lean}}$), so that the Φ_{MDL} vacuum mode count is $3L$, not L^3. The quantum stability of this count — i.e., whether loop corrections restore the L^3 scaling and overwhelm the constant — is addressed in Section 3.6; the answer is no.

The full coefficient combines the three spatial tapes, the proper-time rate τ, and the spacetime dimension $D = 4$:

$$\rho_\Lambda = \frac{N_{\mathrm{spatial}} \cdot \tau}{D^2} M_{\mathrm{Pl}}^2 H_0^2 = \frac{N_{\mathrm{spatial}}^2}{D^2\,|\mathbb{Z}_7|} M_{\mathrm{Pl}}^2 H_0^2 = \frac{3 \cdot (3/7)}{16} M_{\mathrm{Pl}}^2 H_0^2 = \frac{9}{112} M_{\mathrm{Pl}}^2 H_0^2, \tag{6}$$

using $\tau = N_{\mathrm{spatial}}/|\mathbb{Z}_7| = 3/7$ from (3) (`voxel_cc_coefficient`; $\mathbf{A/D}$). The coefficient is fixed entirely by the CMCA architecture — three tapes, seven $\mathbb{Z}_7$ classes, four spacetime dimensions — with no adjustable input. Equivalently, through the Gorard coefficient (4),

$$\frac{9}{112} = 2\, C_{\mathrm{Gorard}}\, \tau = 2 \cdot \frac{3}{32} \cdot \frac{3}{7}, \tag{7}$$

exhibiting the shared $D^2 = 16$ structure (`voxel_coeff_eq_two_c_gorard_tau`; $\mathbf{A}_{\mathrm{Lean}}$).

3.3 Dark-energy fraction, route 1: PSC epoch selection

The first determination of Ω_Λ comes from the residual structural complexity of the substrate under Perfect Self-Containment. The PSC Reflexive Closure axiom requires the substrate to encode its own description; the resulting self-reference is undecidable, and the residual undecidable content sets a non-zero floor on the energy density. Quantitatively, the residual complexity per Hubble time is

$$D_{\rm res} = \frac{\ln 2}{\pi}\, L_{\rm model}\, H_0^2, \qquad L_{\rm model} = \log_2 \frac{2000}{3}, \tag{8}$$

where $L_{\rm model}$ is the binary description length of the gauge-sector $\mathbb{Z}_7$ orbit, which carries 2000/3 orbit classes [14] (`gauge_spectrum_total`, `lmodel_numerator_coprime_denominator`; $\mathbf{A}_{\rm Lean}$). The orbit count $2000/3 = D^2 N_{\rm fam}^{N_{\rm gen}}/N_{\rm gen}$, where $D = N_{\rm gen} + 1 = 4$ follows from the Dimensional Protocol Principle ($\mathbf{A}_{\rm Lean}$, `dimensional_protocol_principle_master`, [21]) and $N_{\rm fam} = 5$ from the $\mathbb{Z}_5$ orbit structure, is derivable from the three-tape CMCA architecture with no external inputs (**A/D**). The dark-energy density is $\rho_\Lambda = D_{\rm res}\, M_{\rm Pl}^2$, and the dark-energy fraction $\Omega_\Lambda = D_{\rm res}/(3H_0^2)$ is purely structural:

$$\boxed{\Omega_\Lambda = \frac{\ln 2}{3\pi}\, \log_2 \frac{2000}{3} = 0.6899.} \tag{9}$$

The residual is strictly positive (`d_res_pos`; **A/D**), Ω_Λ is fully determined by $D_{\rm res}$ (`omega_lambda_from_d_res`, `d_res_determines_omega_lambda`; **A/D**), and the master selection theorem is `psp_epoch_selection_master` ($\mathbf{A}_{\rm Lean}$, zero `sorry`). The value (9) agrees with Planck 2018, $\Omega_\Lambda = 0.6847 \pm 0.0073$ [1], at 0.71σ, with zero free parameters.

3.4 Dark-energy fraction, route 2: holographic degrees of freedom

The second determination follows directly from the energy density (6). Normalizing to the critical density $\rho_{\rm crit} = 3H_0^2 M_{\rm Pl}^2/(8\pi)$,

$$\Omega_\Lambda = \frac{\rho_\Lambda}{\rho_{\rm crit}} = \frac{9}{112} \cdot \frac{8\pi}{3} = \frac{3\pi}{14} = 0.6732, \tag{10}$$

machine-certified as `omega_lambda_from_temporal_voxel` and, through the Gorard coefficient, `omega_lambda_via_gorard` (**A/D**). The structure collapses to the remarkably compact form

$$\Omega_\Lambda = \tau \cdot \frac{\pi}{2} = \frac{N_{\rm spatial}}{|\mathbb{Z}_7|} \cdot \frac{\pi}{2} = \frac{3}{7} \cdot \frac{\pi}{2} = \frac{3\pi}{14} \approx 0.673 \qquad (\mathbf{A/D}). \tag{11}$$

This value agrees with Planck 2018 at -2.28% (-2.8σ). It is the geometric/structural leading-order estimate: the idealization $\tau = N_{\rm spatial}/|\mathbb{Z}_7|$ enters directly, whereas the $D_{\rm res}$ route (9) carries the exact MDL boundary count.

3.5 Two independent predictions and the irreducible range

GTE thus provides a *prediction band* $\Omega_\Lambda \in [3\pi/14,\, D_{\rm res}] = [0.673,\, 0.690]$ from two structurally independent mechanisms. The PSC route (Equation (9)) lands at $+0.2\sigma$ from Planck; the holographic route (Equation (11)) lands at -2.8σ. Together they bracket the observed 0.6889 from below and above. The two routes are independent: (9) arises from PSC undecidability and orbit counting, while (10) arises from holographic mode counting and the proper-time rate. They use no common calibration, yet they bracket the observed value (Table 2). The generation count $N_{\rm gen} = 3$ is the unique integer for which both structural routes simultaneously bracket the Planck 2018 value; for all other $N \in \{1, 2, 4, 5, \ldots\}$ the two routes either both exceed or both fall below the observed Ω_Λ (**A/D**).

The 2.42% spread between the two formulas is irreducible. At the formula level, the two constants are provably distinct transcendentals — the logarithm of a rational in (9) versus π in (10) — so no

Route	**Mechanism**	Ω_Λ	**vs Planck**
PSC reflexive closure	undecidability + $\mathbb{Z}_7$ orbit count	0.6899	$+0.2\sigma$
Holographic DOF	three-tape mode count + proper time	0.6732	-2.8σ
Planck 2018	observation	0.6889	—

Table 2: Two structurally independent derivations of the dark-energy fraction bracket the observed value. Their mutual spread is 2.42%.

exact common ancestor exists. At the level of physical effects, every candidate correction to τ (from the neutrino sector, the dark-sector vacuum, and the special-relativistically measured proper-time rate) was tested and fails the null suite, lying in the wrong direction or with the wrong magnitude. The framework therefore makes an irreducible prediction *range*

$$\Omega_\Lambda \in \left[\frac{3\pi}{14},\ D_{\mathrm{res}}\right] = [0.673,\ 0.690], \tag{12}$$

which brackets the observed 0.6889. Two independent structural counts agreeing to 2.4% and bracketing the observation is a non-trivial internal consistency check.

To see why the gap is irreducible, note that the two formulas evaluate different mathematical functions at the same physical point:

$$\Omega_\Lambda^{\mathrm{PSC}} = \frac{\ln 2}{3\pi}\,\log_2\!\left(\frac{2000}{3}\right), \tag{13}$$

$$\Omega_\Lambda^{\mathrm{voxel}} = \frac{3\pi}{14}, \tag{14}$$

where the first involves $\ln 2$ and $\log_2(2000/3)$ (a rational logarithm) and the second involves only π. These belong to different transcendental degrees: $\ln 2$ is a value of the logarithm function at a rational point, while π is transcendental of a different kind. By the Lindemann–Weierstrass theorem (and standard transcendence theory), no algebraic relation of bounded degree connects these two constants, so no GTE arithmetic identity can close the gap. The 2.42% is not a deficit to be explained away; it is a diagnostic of genuine structural independence between the two derivation routes, confirming that the bracketing is a real cross-check rather than a tautological rearrangement of the same formula.

Key Result: Cosmological Constant at Three Levels

Magnitude (A/D): $\rho_\Lambda = \dfrac{N_{\mathrm{spatial}}^2}{D^2\,|\mathbb{Z}_7|}\,M_{\mathrm{Pl}}^2 H_0^2 = \dfrac{9}{112}\,M_{\mathrm{Pl}}^2 H_0^2$, in units where the coefficient is fixed entirely by three tapes, seven $\mathbb{Z}_7$ classes, and four spacetime dimensions.

Value, route 1 — PSC reflexive closure (A/D):

$$\Omega_\Lambda = \frac{\ln 2}{3\pi}\log_2\frac{2000}{3} = 0.6899, \quad +0.71\sigma \text{ from Planck}$$

Value, route 2 — holographic mode count (A/D):

$$\Omega_\Lambda = \tau\cdot\frac{\pi}{2} = \frac{3\pi}{14} = 0.6732, \quad -2.8\sigma \text{ from Planck}$$

Irreducible prediction range: $\Omega_\Lambda \in [0.673, 0.690]$ — brackets the observed 0.6889; the two constants are provably distinct transcendentals so no exact common formula exists.

Quantum protection (**A/D**-partial): holographic one-loop/tree ratio 1.22×10^{-42} vs. the standard 1.66×10^{40}; suppression $3H_0^2/M_{\text{kink}}^2 \approx 7.4 \times 10^{-83}$ (geometric origin, not supersymmetric cancellation).

3.6 Quantum protection: holographic non-renormalization

A constant fixed at tree level is only physical if quantum loops do not overwhelm it. The holographic mode count answers this question. In standard quantum field theory, the one-loop correction to the vacuum energy relative to the tree-level dark-energy density is

$$\frac{\delta\rho_{\text{loop}}}{\rho_{\text{tree}}} \sim \frac{g^2}{16\pi^2}\frac{M_{\text{kink}}^4}{\rho_{\text{tree}}} \approx 1.66 \times 10^{40}, \tag{15}$$

which is the hierarchy problem in sharp form. In the CMCA, the vacuum carries $3L$ modes rather than L^3, so one power of the ultraviolet cutoff M_{kink}^2 is replaced by $3H_0^2$:

$$\frac{\delta\rho_{\text{loop}}}{\rho_{\text{tree}}} \sim \frac{g^2}{16\pi^2}\frac{M_{\text{kink}}^2 \cdot 3H_0^2}{\rho_{\text{tree}}} \approx 1.22 \times 10^{-42}. \tag{16}$$

Equivalently, substituting $\rho_{\text{tree}} = \frac{9}{112} M_{\text{Pl}}^2 H_0^2$, the H_0 factors cancel and the ratio takes the manifestly epoch-independent form

$$\frac{\delta\rho_{\text{loop}}}{\rho_{\text{tree}}} = \frac{g_{\text{fund}}^2}{16\pi^2} \cdot \frac{112}{3} \cdot \left(\frac{m_{\text{kink}}}{M_{\text{Pl}}}\right)^2 = 1.22 \times 10^{-42} \tag{17}$$

which is *independent of* H_0: it depends only on the GTE mass ratio $m_{\text{kink}}/M_{\text{Pl}}$, with the Hubble scale cancelling identically. This scale-invariance is certified by `holographic_nrt_loop_tree_ratio_h0_independent` (zero sorry; `TemporalVoxelCC.lean`). The NRT mechanism is thus a structural property of the GTE mass hierarchy, valid at any cosmological epoch.

The suppression relative to the standard estimate is

$$\frac{3H_0^2}{M_{\text{kink}}^2} \approx 7.4 \times 10^{-83}, \tag{18}$$

computed with $g = 49/512$, $M_{\text{kink}} = 0.29010$ GeV, and $H_0 = 67.4$ km/s/Mpc (`holographic_mode_count_ratio`, `holographic_voxel_scaling`, `holographic_identification_voxel_cc`; **A/D**-partial, zero `sorry`). Quantum loops cannot overwhelm the constant: the holographic counting suppresses them by more than forty orders of magnitude relative to the standard field-theory estimate. This is a non-renormalization mechanism of geometric origin, distinct from supersymmetry, which we contrast in Section 8.2.

Key Result: Holographic Non-Renormalization

The holographic mode count ($3L$ rather than L^3) suppresses one-loop corrections to the cosmological constant:

$$\frac{\delta\rho_{\text{loop}}}{\rho_{\text{tree}}} \sim \frac{g^2}{16\pi^2}\frac{M_{\text{kink}}^2 \cdot 3H_0^2}{\rho_{\text{tree}}} \approx 1.22 \times 10^{-42}$$

compared to the standard field-theory estimate 1.66×10^{40}. The suppression factor is $3H_0^2/M_{\text{kink}}^2 \approx 7.4 \times 10^{-83}$: purely geometric, requiring no superpartner spectrum. (**A/D**-partial.)

This result is **A/D**-partial: the phenomenological question is answered, but the complete Φ_{MDL} path-integral derivation, showing that the mode sum is exactly $3L$ and not a divergent L^3, is a future result (Section 8.3).

The two suppressions address different faces of the cosmological constant problem. The holographic mode-count suppression derived in §2.2 resolves the magnitude problem: the naive one-loop estimate $\rho_{\rm vac}^{\rm naive} \sim M_{\rm Pl}^4$ is reduced by the holographic mode fraction $3H_0^2/M_{\rm Pl}^2 \approx 10^{-122}$ to the observed scale $\rho_\Lambda \sim M_{\rm Pl}^2 H_0^2$. The loop-stability suppression derived here ($\sim 10^{-42}$) then acts on top of this reduced estimate, ensuring that quantum corrections do not destabilize the classical $\Lambda = 0$ solution into a large residual. Together, these provide a complete resolution: the magnitude is removed by holographic counting, and the stability of that reduction is established by the tree-to-loop ratio. The traditional Zeldovich formulation of the CC problem ($\Lambda^{\rm bare}$ cancelling loop corrections to 120 decimal places) does not arise in GTE because the holographic counting removes the need for such a cancellation: the physical CC is not set by a near-cancellation of large numbers but by a structural mode count that yields $\rho_\Lambda \sim M_{\rm Pl}^2 H_0^2$ directly. The open question (OQ-CC-QUANTUM) is why the two independent structural counts (PSC epoch route and holographic mode route) bracket the observed value at the 2.4% level.

Sector grading and fermionic excess. The $\mathbb{Z}_2^*$ multiplicative-order structure of the winding-sector alphabet certifies a $(-1)^F$ grading: fermionic sectors $\{2, 4, 6\}$ and bosonic sectors $\{0, 3\}$, as proved in [21] (`gte_fermionic_sectors_get_minus_phase`, $\mathbf{A}_{\rm Lean}$ + 1 topological axiom). Among the PSC-admissible spectrum this gives $\mathrm{Str}(1) = n_B - n_F = 2 - 3 = -1$ (net-fermionic) and $\mathrm{Str}(\mathrm{DOF}) = 13 - 84 = -71$: 13 bosonic degrees of freedom from sectors $\{0, 3\}$ and 84 fermionic from the up quarks, charged leptons, and down quarks of the Standard Model mapped onto sectors $\{2, 4, 6\}$. The graded one-loop contribution is

$$-\mathrm{Str}(\mathrm{DOF}) \cdot \frac{m_{\rm kink}^4}{32\pi^2} \;\approx\; -1.59 \times 10^{-3}\ \mathrm{GeV}^4,$$

which has the correct negative sign but is $\approx 5 \times 10^{43}$ times larger than $\rho_{\rm obs}$. This magnitude-43-order mismatch is the standard cosmological constant problem in QFT form: the holographic non-renormalization (Section 3.6) suppresses this sector one-loop contribution (as well as the naive mode one-loop) to complete negligibility, as derived below. The physically dominant result is from the holographic term alone: $\rho_\Lambda = 2.48 \times 10^{-47}$ GeV^4 (Equation (11)), which already reaches 88.6% of $\rho_{\rm obs}$ without any sector correction. The prior assumption that $\Phi_{\rm MDL}$ is purely bosonic is incorrect: the GTE vacuum carries a certified net-fermionic grading ($\mathrm{Str}(\mathrm{DOF}) = -71$) with the correct sign for partial cancellation.

Holographic suppression of the sector DOF loop. When the holographic mode count ($3L$ not L^3) is applied consistently to the graded DOF, the sector-graded one-loop contribution becomes

$$\rho_{\rm graded}^{\rm holo} = \tfrac{1}{2}\,\mathrm{Str}(\mathrm{DOF})\, m_{\rm kink}\, \frac{3H_0^2}{M_{\rm Pl}^2} \approx -5 \times 10^{-102}\ \mathrm{GeV}^4, \tag{19}$$

which is $\approx 10^{-55}\,\rho_{\rm obs}$ — completely negligible. The holographic mode count therefore suppresses *both* the naive mode loop (Section 3.6) and the sector-graded loop, leaving only the holographic CC $\rho_\Lambda = (9/112) M_{\rm Pl}^2 H_0^2$ as the surviving contribution. This resolves the apparent tension: the $\mathrm{Str}(\mathrm{DOF}) = -71$ is a real physical grading, but the holographic bound replaces (not supplements) the QFT one-loop calculation, so the standard $m_{\rm kink}^4$ sector loop does not appear as a separate term.

3.7 The equation of state: $w = -1$ from a boundary condition

PSC epoch selection is a boundary condition on the cosmic history, not a dynamical field equation. The dark-energy density is fixed at its PSC-selected value, so $\mathrm{d}\rho_\Lambda/\mathrm{d}t = 0$ by construction. Standard energy conservation, $\dot{\rho} + 3H(\rho + p) = 0$, then forces $p = -\rho_\Lambda$, i.e.

$$w \equiv \frac{p}{\rho} = -1. \tag{20}$$

This is the equation of state of a true cosmological constant and the source of accelerated expansion. It is the decisive structural difference from holographic dark energy [3, 8], whose density tracks the horizon and therefore predicts $w = 0$ and no acceleration. Table 3 summarizes the comparison. A Lean formalization of the $w = -1$ derivation from `psp_epoch_selection_master` is a target for future work; the argument here is **A/D**.

Property	**CKN holographic DE**	**GTE**
$\rho_\Lambda \sim M_{\text{Pl}}^2 H_0^2$	yes	yes (holographic suppression)
Coefficient	undetermined	9/112
Equation of state	$w \approx 0$ (fatal)	$w = -1$ (PSC boundary)
Accelerated expansion	no	yes

Table 3: The GTE construction shares the $M_{\text{Pl}}^2 H_0^2$ scaling of Cohen–Kaplan–Nelson holographic dark energy but fixes both the coefficient and $w = -1$.

3.8 Evasion of Weinberg's no-go theorem

Weinberg's no-go theorem [23] excludes any adjustment of the cosmological constant by a scalar field coupled to gravity through a local effective action. The GTE mechanism evades it structurally rather than by fine-tuning: the selection of the residual value is performed by the PSC adjudication process, which is non-effective (it cannot be reduced to a total Turing-computable rule on the self-referential fragment) and therefore lies outside the class of local effective-field-theory adjustments to which the no-go applies. The mechanism is not a new scalar field or extra dimension; it is a property of the PSC undecidability structure of the substrate. The complete chain from PSC halting-undecidability to a strictly positive cosmological constant is machine-certified: `incompleteness_implies_nonzero_omega_lambda` (zero `sorry`).

4 The CMB Spectral Tilt

4.1 Binary entropy and the holographic running rate

The scalar spectral index of the primordial power spectrum measures the slight scale dependence of the perturbation amplitude. In the GTE framework the perturbation spectrum is governed by the binary ($\mathbb{Z}_2$) sublayer of the CMCA, which carries the binary entropy $H(\mathbb{Z}_2) = \ln 2$ per holographic screen cell.

The running rate of the Newton coupling across scales is fixed by the Weyl mode count on the spatial three-sphere S^3 at the Planck scale. The Weyl law gives the number of modes below wavenumber k as $N(k) = k^3/3$ in units where the radius is one (`weyl_mode_count_eq`; $\mathbf{A}_{\text{Lean}}$), so the logarithmic derivative at the Planck scale is

$$\left.\frac{\mathrm{d}N}{\mathrm{d}\ln k}\right|_{k_{\text{Pl}}} = \frac{\mathrm{Vol}(S^3)}{2\pi^2} = 1, \tag{21}$$

using $\mathrm{Vol}(S^3) = 2\pi^2$ — an exact algebraic cancellation (`weyl_log_deriv_at_planck`, `weyl_algebraic_miracle`; $\mathbf{A}_{\mathrm{Lean}}$). The classical discrete renormalization-group rate is the product of the binary entropy and this Weyl factor:

$$\frac{\mathrm{d}K_{\mathrm{local}}}{\mathrm{d}\ln k} = H(\mathbb{Z}_2) \cdot 1 = \ln 2, \tag{22}$$

(`classical_discrete_rg_entropy_rate`; $\mathbf{A}_{\mathrm{Lean}}$), and the Newton-coupling running rate is

$$\beta_G = \frac{H(\mathbb{Z}_2)}{\mathrm{Vol}(S^3)} = \frac{\ln 2}{2\pi^2} = 0.03512, \tag{23}$$

(`beta_g_from_classical_rg`, `beta_g_from_weyl_law`; $\mathbf{A}_{\mathrm{Lean}}$).

4.2 The spectral index

The CMB scalar spectral index follows from the running rate (23) through the standard pump relation $n_s = 1 - (3/2)\beta_G \cdot (4/3) = 1 - \beta_G$:

$$\boxed{n_s = 1 - \frac{\ln 2}{2\pi^2} = 0.96488.} \tag{24}$$

This is machine-certified as `n_s_formula`, with the qualitative red tilt $n_s < 1$ certified separately (`n_s_less_than_one`), and the identification of the physical spectral index with the GTE prediction (`z2_eft_predicts_cmb_tilt`, `cmb_spectral_index_equals_gte_prediction`). The full chain comprises fourteen theorems in the CMB spectral-tilt module, all with zero `sorry` and zero axioms. The value (24) agrees with Planck 2018, $n_s = 0.9649 \pm 0.0042$ [1], at 0.004σ — the most precise agreement between a GTE prediction and CMB data.

Key Result: CMB Scalar Spectral Index

$$n_s = 1 - \frac{\ln 2}{2\pi^2} = 0.96488, \qquad 0.004\sigma \text{ from Planck 2018}$$

Derived from the binary $\mathbb{Z}_2$ entropy $H(\mathbb{Z}_2) = \ln 2$ and the algebraic cancellation $\mathrm{Vol}(S^3)/(2\pi^2) = 1$. Certified by fourteen zero-`sorry` Lean theorems ($\mathbf{A}_{\mathrm{Lean}}$).

4.3 The $\mathbb{Z}_2/\mathbb{Z}_7$ separation

A subtle point is why the running rate is governed by the binary entropy $H(\mathbb{Z}_2) = \ln 2$ rather than by the full $\mathbb{Z}_7$ entropy $H(\Phi_{\mathrm{MDL}})$. The reason is superselection: the Doplicher–Haag–Roberts theorem [5] forbids coherent superpositions across the $\mathbb{Z}_7$ winding superselection sectors, and the flat metric preserves the $\mathbb{Z}_7$ superselection structure (`z7_superselection_preserved_by_flat_metric`; $\mathbf{A}_{\mathrm{Lean}}$). The perturbation spectrum therefore couples only to the binary parity sublayer, whose entropy is $\ln 2$. The $\mathbb{Z}_7$ route, by contrast, predicts $n_s(\mathbb{Z}_7) = -15.11\sigma$ and is ruled out. The physical identification of the binary effective field theory with a classical scalar field on a compact S^4 is $\mathbf{A/D}$; its full field-theoretic derivation is an open problem (Section 8.3).

5 Gravitational Physics

5.1 The Gorard discrete-smooth bridge and the Newton constant

The Newton constant is normalized through the discrete Ollivier–Ricci curvature of the CMCA causal graph in the coarse-graining limit of Gorard [6]. The curvature carried by a Standard-Model glider is the rational invariant

$$\kappa_{\mathrm{SD}} = \frac{10}{13}, \tag{25}$$

(`kappa_SD_eq_10_13`; $\mathbf{A}_{\mathrm{Lean}}$), and the discrete Bianchi identity holds: $\sum \kappa_{\mathrm{OR}} = 0$ over every closed spacelike loop. The complete chain of twenty discrete-to-smooth implications, mapping the graph curvature to the vacuum Einstein equations, is bundled as `gorard_chain_catAL_master_bundle` ($\mathbf{A}_{\mathrm{Lean}}$, zero `sorry`). The coarse-graining coefficient $C_{\mathrm{Gorard}} = 3/32$ of (4) fixes the Newton-constant normalization. The same $D^2 = 16$ that appears here also appears in the dark-energy coefficient (7), so the gravitational and cosmological normalizations share a single origin.

5.2 The strong-field UV bound

The $\mathbb{Z}_7$ potential is bounded above. Its maximum value is

$$V_{\mathrm{max}} = \frac{2m^2}{49}, \tag{26}$$

attained at $\theta = \pi/7$ (`strong_field_uv_bound_catad`, `z7_potential_max_value`, `vZ7_le_max`; **A/D**). Beyond this value the GTE effective field theory breaks down and gravity becomes strongly coupled. The bound implies a natural cutoff on primordial density perturbations at the Planck-kink mass scale and a corresponding quantitative constraint on primordial black hole formation.

5.3 Hawking radiation as kink emission

The BPS kink mass saturates the strong-field bound:

$$M_{\mathrm{kink}} = \frac{8}{49}\, m_\tau, \qquad M_{\mathrm{kink}}^{\mathrm{crit}} = 290.10\ \mathrm{MeV}, \tag{27}$$

(`kink_bps_mass_formula`, `kink_crit_mass_formula`; $\mathbf{A}_{\mathrm{Lean}}$). Near this critical mass, an evaporating black hole emits kinks — topological excitations of the Φ_{MDL} field — rather than purely thermal radiation. The emission is Bose-enhanced at the Hawking temperature (`kink_over_hawking_temp_ratio`, `kink_bose_enhanced_at_Mcrit`; $\mathbf{A}_{\mathrm{Lean}}$), and the complete emission profile is bundled as `hawking_kink_emission_catad` (**A/D**). The cosmological implication is a distinctive non-thermal spectral feature near 290 MeV in the evaporation spectrum of primordial black holes, absent from standard Hawking radiation [7], together with a kink relic density and a stochastic gravitational-wave background whose quantitative spectrum is an open problem.

6 The Neutrino Sector and Dark Matter

6.1 Neutrino masses as a cosmological observable

The neutrino mass sum is a cosmological observable: the cosmic microwave background and large-scale structure constrain Σm_ν through its effect on structure growth. The GTE seesaw with right-handed mass $M_R = 1.11 \times 10^{13}$ GeV gives a lightest neutrino mass $m_{\nu_1} = 0.679$ meV and a sum

$$\Sigma m_\nu = 59.4\ \mathrm{meV}, \tag{28}$$

(**A**) [17], comfortably below the Planck bound $\Sigma m_\nu < 120$ meV by a factor of two and within reach of the next generation of CMB and large-scale-structure surveys (CMB-S4, Euclid). A measured value inconsistent with ~ 60 meV would falsify the prediction.

6.2 Leptogenesis

The same neutrino sector supports thermal leptogenesis. The GTE efficiency factor is $K_1 = 15.93$ (**A**), and the required CP asymmetry $\varepsilon_1 \approx 1.8 \times 10^{-5}$ lies well below the Davidson–Ibarra bound [4], so the observed baryon asymmetry is feasible without fine-tuning; 77% of order-one Yukawa textures are consistent (**B**naturalness).

6.3 Dark matter

The GTE dark sector [15] supplies a stable mirror-braid candidate. The relic-density calculation and CMB constraints for the specific dark lepton masses are not worked out here and are noted as open (Section 8.3).

7 The CKM CP Phase from an S_3 Subgroup Chain

The CKM CP-violating phase enters the cosmological story as an input to the baryon-asymmetry chain. In the GTE framework it is fixed by the residual-symmetry structure of the three families under the symmetric group S_3. The phase is the distance from maximal CP violation $\pi/2$ minus the ratio of the down-sector residual symmetry to the complementary sectors:

$$\delta_{CP} = \frac{\pi}{2} - \frac{|H_{\text{down}}|}{|H_{\text{lep}}| + |H_{\text{up}}|} = \frac{\pi}{2} - \frac{3}{6+2} = \frac{\pi}{2} - \frac{3}{8} = 68.51^\circ, \tag{29}$$

where the S_3 subgroup chain selects $|H_{\text{down}}| = 3$ (the unique normal $A_3 \cong \mathbb{Z}_3$ subgroup), $|H_{\text{lep}}| = 6$ (the full S_3), and $|H_{\text{up}}| = 2$ (a $\mathbb{Z}_2$ reflection). This agrees with the PDG value 68.53° to 0.017% (`ckm_cp_phase_formula`, `cp_phase_reduction_ratio`; **A**, Lean-certified identity). Four independently motivated variants of the ratio each lie $\geq 10\%$ from the PDG value, so (29) is not a fit (`cp_phase_ratio_distinct_from_nulls`).

Key Result: CKM CP Phase

$$\delta_{CP} = \frac{\pi}{2} - \frac{3}{8} = 68.51^\circ, \qquad 0.017\% \text{ from PDG } 68.53^\circ$$

Fixed by the S_3 subgroup chain: $|H_{\text{down}}| = 3$, $|H_{\text{lep}}| = 6$, $|H_{\text{up}}| = 2$. Four null variants lie $\geq$10% from PDG; no free parameter is adjusted. (**A**, Lean-certified identity.)

Quark Yukawa cone angles. The same S_3 subgroup chain predicts cone-angle parameters for the quark sectors. Writing $\theta_{\text{sector}} = |H_{\text{sector}}|/N_c^3$ where $N_c = 3$, the down-quark sector ($H = A_3 \cong \mathbb{Z}_3$, $|H| = 3$) gives $\theta_{\text{down}} = 3/27 = 1/9$, and the up-quark sector ($H = \mathbb{Z}_2$, $|H| = 2$) gives $\theta_{\text{up}} = 2/27$. The lepton sector gives the Lean-certified $\theta_{\text{lep}} = 6/27 = 2/9$ ($\mathbf{A}_{\text{Lean}}$). Fitted to PDG quark masses, θ_{down} and θ_{up} lie 0.84% and 0.43% from the observed values, respectively (**A**). These are open predictions of the GTE Koide mechanism for the quark sector.

The Jarlskog invariant follows as $J = 3.02 \times 10^{-5}$, 5% from the PDG value 3.18×10^{-5}; the residual is consistent with the 3.6% gap in the Wolfenstein A parameter, $A = \sin(\pi/3)$ (within 2σ of PDG) (`ckm_A_parameter_sin2`; **A**). Closing the A parameter would close J.

Wolfenstein A parameter. The Wolfenstein A parameter satisfies $A \approx \sin(\pi/|H_{\text{down}}|) = \sin(\pi/3) = \sqrt{3}/2 \approx 0.866$, lying 3.6% above the PDG value 0.836. The deviation falls within the experimental tension between inclusive and exclusive $|V_{cb}|$ determinations, so $\sin(\pi/3)$ is consistent at the $\sim 2\sigma$ level (**A**). Resolving the inclusive/exclusive tension would sharpen this test directly.

8 Discussion

8.1 Comparison with holographic dark energy

The closest prior work to the present cosmological-constant mechanism is the holographic dark-energy programme of Cohen, Kaplan, and Nelson [3] and Hsu [8]. All three share the key structural insight that gravity correlates ultraviolet and infrared scales and thereby depletes the effective number of field-theory modes, yielding $\rho_\Lambda \sim M_{\rm Pl}^2 H_0^2$. The GTE construction differs in two respects. It fixes the numerical coefficient 9/112 from the CMCA architecture, whereas the holographic bound leaves it free. And it predicts $w = -1$ from the PSC boundary condition, whereas the holographic density tracks the horizon and predicts $w = 0$ — the obstruction that prevents holographic dark energy from producing accelerated expansion [8]. The GTE dark-energy density is a constant of the cosmic evolution by self-containment, not a horizon-tracking quantity, which is precisely why the equation of state is that of a true cosmological constant.

A further distinction from both CKN and Bekenstein–Hawking holography is the *dimension* of holographic encoding. Standard BH holography encodes a 3D volume on its 2D surface, giving mode suppression $L^2/L^3 = 1/L$ and density scaling $M_{\rm Pl}^3 H_0$ — still $\sim 10^{61}$ times larger than observed. The three-tape CMCA encodes the 3D spacetime in *one*-dimensional tapes, giving the stronger suppression $3L/L^3 = 3/L^2$ and the observed $M_{\rm Pl}^2 H_0^2$ scaling (Equation (2)). The CMCA is thus more holographically compressed than a black hole, and this additional compression is precisely the mechanism that bridges BH holography to the correct cosmological-constant magnitude.

The holographic L^1 bound studied here (Section 3.2) may also have implications for particle-mass predictions: the Information Mass Transformer of the GTE verifier [9] currently uses the Bekenstein L^2 bound; upgrading to the tighter CMCA L^1 bound is a direction for follow-on work.

8.2 Holographic non-renormalization versus supersymmetry

Supersymmetry protects the vacuum energy by cancellation between bosonic and fermionic loops, requiring degenerate superpartners — a structure not observed at accessible energies. The holographic non-renormalization of Section 3.6 is different in kind: the loop suppression $3H_0^2/M_{\rm kink}^2$ is geometric, arising from the boundary mode count $3L$ rather than the bulk count L^3, and requires no partner spectrum. The $\approx 10^{-42}$ suppression is a property of the holographic architecture, not an algebraic cancellation between species.

8.3 Open problems

Open Problem 1 (Full path-integral non-renormalization). The holographic mode count suppresses sector DOF loops to $\rho^{\rm holo}_{\rm graded} \approx -5 \times 10^{-102}$ GeV^4 (*shown above, Section 3.6*), confirming that the holographic bound replaces the QFT vacuum-energy computation entirely. The remaining open question is a formal path-integral proof of this replacement (gated on OQ-QG-1): a complete $\Phi_{\rm MDL}$ path-integral computation showing that the mode sum is exactly $3L$ and not an ultraviolet-divergent L^3 would upgrade this result from **A/D**-partial to **A/D**.

Open Problem 2 (H_0 from the baryon asymmetry). H_0 enters the $D_{\rm res}$ formula (9) as an observational input. A candidate first-principles chain runs from the GTE quark Koide parameters through the Jarlskog invariant and δ_{CP} (29) to the baryon asymmetry η_B, the baryon density, and hence H_0 via the Friedmann equation at matter–radiation equality. The Jarlskog and CP-phase links are established (Section 7); the remaining open step is the derivation of η_B from the Sakharov conditions using the GTE non-equilibrium phase transition. If this chain closes, Ω_Λ becomes fully zero-input.

Open Problem 3 (CMB tilt field-theory identification). The algebraic chain giving $n_s = 1 - \ln 2/(2\pi^2)$ is $\mathbf{A}_{\text{Lean}}$. The physical identification of the binary effective field theory with a classical scalar field on a compact S^4 governing the perturbation spectrum is $\mathbf{A/D}$ and not yet $\mathbf{A}_{\text{Lean}}$.

Open Problem 4 (Wolfenstein A parameter and the Jarlskog residual). The 3.6% gap in $A = \sin(\pi/3)$ accounts for most of the 5% residual in the Jarlskog invariant. A mechanism deriving the A parameter from the Φ_{MDL} Yukawa structure would close both.

Open Problem 5 (Gravitational-wave background and dark-matter relic density). The kink emission of Section 5 produces a stochastic gravitational-wave background from evaporating primordial black holes; its frequency peak and amplitude are not yet computed. The relic density of the dark sector is likewise open.

8.4 Falsifiability

Table 4 states the falsifiability profile. Every prediction is sharp and testable; several are within reach of near-future experiments.

Prediction	**Observable**	**Current bound**	**How to falsify**
$\Omega_\Lambda \in [0.673, 0.690]$	CMB + BAO	0.6847 ± 0.0073	future CMB shifting Ω_Λ outside the range
$n_s = 0.96488$	CMB power spectrum	0.9649 ± 0.0042	any future shift of n_s by > 0.01
$\Sigma m_\nu = 59.4$ meV	CMB-S4 + Euclid	< 120 meV	direct measurement excluding ~ 60 meV
kink emission near $M_{\text{kink}}^{\text{crit}}$	PBH spectrum	none yet	null of a non-thermal bump near 290 MeV
$V_{\max} = 2m^2/49$	primordial GW/PBH	unconstrained	primordial spectra inconsistent with the cutoff

Table 4: Falsifiability profile of the GTE cosmological predictions.

9 Conclusions

We have assembled the cosmological predictions of the GTE/Φ_{MDL} framework and derived them from zero-parameter first principles. The cosmological constant is addressed at three levels. Its *value* is fixed by two structurally independent counts — the PSC reflexive-closure route $\Omega_\Lambda = 0.6899$ and the holographic route $\Omega_\Lambda = 3\pi/14 = 0.6732$ — that bracket the observed 0.6889 and define an irreducible prediction range. Its *magnitude* follows from the holographic mode count, $\rho_\Lambda = (9/112)\, M_{\text{Pl}}^2 H_0^2$, with the equation of state $w = -1$ fixed by the PSC boundary condition rather than by dynamics, evading both Weinberg's no-go theorem and the Cohen–Kaplan–Nelson $w = 0$ obstruction. Its *quantum protection* follows from the same holographic counting, which suppresses one-loop corrections by $\approx 10^{-42}$ relative to the standard field-theory estimate. The CMB scalar spectral index $n_s = 1 - \ln 2/(2\pi^2) = 0.96488$ agrees with Planck 2018 at 0.004σ, certified by fourteen zero-`sorry` theorems. The Newton-constant normalization, the strong-field bound, Hawking kink emission, the neutrino sum, and the CKM CP phase follow from the same substrate. All central results are machine-checked in Lean 4. The remaining open problems — the formal path-integral proof of holographic mode-sum replacement (sector DOF loops are shown to be negligible at $\sim 10^{-102}$ GeV^4), the derivation of H_0 from the baryon asymmetry, and the field-theory identification of the CMB tilt — are stated precisely, and the predictions are sharply falsifiable.

Unification through the GTE polynomial. Every result in this paper is a consequence of a single combinatorial object: the Generative Triple Evolution polynomial over GF(7), specified in full by 19 bits. From this one polynomial the CMCA architecture is fixed, which determines $N_{\text{spatial}} = 3$, $|\mathbb{Z}_7| = 7$, and $D = 4$, which in turn give $\tau = 3/7$ and the holographic mode count $3L$, which yield $\Omega_\Lambda = 3\pi/14$. The orbit count $|\mathcal{O}_{GTE}| = 2000/3$ determines $\Omega_\Lambda = 0.6899$ by PSC reflexivity. The same polynomial fixes the $\mathbb{Z}_7$ winding structure, giving n_s, G_N, M_{crit}, Σm_ν, and δ_{CP}. The fact that so many cosmological observables — spanning 60 orders of magnitude in energy scale — all trace back to a single 19-bit specification is perhaps the strongest argument for taking the GTE framework seriously as a candidate foundation for physics.

Acknowledgments

The author thanks the open-source Lean 4 and Mathlib communities, whose libraries underpin the machine certifications reported here.

A Lean Certification Inventory

Table 5 lists the theorems cited in this paper, all in the canonical `ugp-lean` library. Unless noted, each has zero `sorry` and zero custom axioms.

Table 5: Lean certification inventory.

Theorem	**Module**	**Status**	**Section**
`psp_epoch_selection_master`	`PSCEpochSelection`	$\mathbf{A}_{\text{Lean}}$	3.3
`incompleteness_implies_nonzero_omega_lambda`	`PSCEpochSelection`	$\mathbf{A}_{\text{Lean}}$	3.3
`omega_lambda_from_d_res`	`PSCEpochSelection`	**A/D**	3.3
`d_res_pos`	`PSCEpochSelection`	**A/D**	3.3
`d_res_determines_omega_lambda`	`PSCEpochSelection`	$\mathbf{A}_{\text{Lean}}$	3.3
`gauge_spectrum_total`	`OQ26Arithmetic`	$\mathbf{A}_{\text{Lean}}$	3.3
`lmodel_numerator_coprime_denominator`	`OQ26Arithmetic`	$\mathbf{A}_{\text{Lean}}$	3.3
`voxel_cc_coefficient`	`TemporalVoxelCC`	**A/D**	3.2
`omega_lambda_from_temporal_voxel`	`TemporalVoxelCC`	**A/D**	3.4
`omega_lambda_via_gorard`	`TemporalVoxelCC`	$\mathbf{A}_{\text{Lean}}$	3.4
`voxel_coeff_eq_two_c_gorard_tau`	`TemporalVoxelCC`	$\mathbf{A}_{\text{Lean}}$	3.2
`c_gorard_eq_n_spatial_over_two_dsq`	`TemporalVoxelCC`	$\mathbf{A}_{\text{Lean}}$	2.2
`holographic_voxel_scaling`	`TemporalVoxelCC`	**A/D**	3.6
`holographic_mode_count_ratio`	`TemporalVoxelCC`	**A/D**	3.6
`holographic_identification_voxel_cc`	`TemporalVoxelCC`	**A/D**	3.6
`holographic_nrt_loop_tree_ratio_h0_independent`	`TemporalVoxelCC`	**A/D**	3.6

Theorem	Module	Status	Section
tau_three_sevenths_from_ether	EtherProperTimeRate	**A/D**	2.2
ether_period_seven	EtherProperTimeRate	**A/D**	2.2
ether_odd_fire_count	EtherProperTimeRate	**A/D**	2.2
phimdl_tau_from_ether	PhiMDLProperTimeBridge	**A/D**	2.2
phimdl_voxel_coeff_from_ether_tau	PhiMDLProperTimeBridge	**A/D**	2.2
phimdl_omega_lambda_from_ether_dynamics	PhiMDLProperTimeBridge	**A/D**	2.2
l1_to_l2_tau_bridge	PhiMDLProperTimeBridge	**A/D**	2.2
n_s_formula	CMBSpectralTilt	$\mathbf{A}_{\text{Lean}}$	4
n_s_less_than_one	CMBSpectralTilt	$\mathbf{A}_{\text{Lean}}$	4
beta_g_from_classical_rg	CMBSpectralTilt	$\mathbf{A}_{\text{Lean}}$	4
beta_g_from_weyl_law	CMBSpectralTilt	$\mathbf{A}_{\text{Lean}}$	4
classical_discrete_rg_entropy_rate	CMBSpectralTilt	$\mathbf{A}_{\text{Lean}}$	4
weyl_log_deriv_at_planck	CMBSpectralTilt	$\mathbf{A}_{\text{Lean}}$	4
weyl_algebraic_miracle	CMBSpectralTilt	$\mathbf{A}_{\text{Lean}}$	4
z2_eft_predicts_cmb_tilt	CMBSpectralTilt	$\mathbf{A}_{\text{Lean}}$	4
z7_superselection_preserved_by_flat_ metric	MinimalCoupling	$\mathbf{A}_{\text{Lean}}$	4
kappa_SD_eq_10_13	GorardRationalFormula	$\mathbf{A}_{\text{Lean}}$	5
gorard_chain_catAL_master_bundle	GorardRationalFormula	$\mathbf{A}_{\text{Lean}}$	5
strong_field_uv_bound_catad	PMDLGravityTheorems	**A/D**	5
z7_potential_max_value	PMDLGravityTheorems	$\mathbf{A}_{\text{Lean}}$	5
kink_bps_mass_formula	PMDLGravityTheorems	$\mathbf{A}_{\text{Lean}}$	5
kink_crit_mass_formula	PMDLGravityTheorems	$\mathbf{A}_{\text{Lean}}$	5
kink_over_hawking_temp_ratio	PMDLGravityTheorems	$\mathbf{A}_{\text{Lean}}$	5
kink_bose_enhanced_at_Mcrit	PMDLGravityTheorems	$\mathbf{A}_{\text{Lean}}$	5
hawking_kink_emission_catad	PMDLGravityTheorems	**A/D**	5
phimdl_tmunu_vacuum_zero	StressEnergyTensor	$\mathbf{A}_{\text{Lean}}$	3.1
z7_vacuum_energy_mass_independent	NRTVacuumEnergy	$\mathbf{A}_{\text{Lean}}$	3.1
ckm_cp_phase_formula	CKMCPPhase	**A**	7
cp_phase_reduction_ratio	CKMCPPhase	**A**	7
ckm_A_parameter_sin2	CKMCPPhase	**A**	7

B Numerical Values and Conversion Factors

- $M_{\text{Pl}} = 1.221 \times 10^{19}$ GeV (Planck mass).

- $H_0 = 67.4$ km/s/Mpc $= 1.44 \times 10^{-42}$ GeV (Planck 2018).
- $\rho_{\rm crit} = 3H_0^2 M_{\rm Pl}^2/(8\pi)$ (critical density).
- $m_\tau = 1776.86$ MeV; $M_{\rm kink}^{\rm crit} = (8/49)m_\tau = 290.10$ MeV.
- $g = 49/512$ (fundamental coupling); $M_R = 1.11 \times 10^{13}$ GeV.
- $N_{\rm spatial} = 3$; $D = 4$; $|\mathbb{Z}_7| = 7$; $\tau = 3/7$.
- $C_{\rm Gorard} = 3/32$; $\kappa_{\rm SD} = 10/13$; CC coefficient $= 9/112$.

C GTE Predictions versus Observation

Table 6 collects every quantitative prediction against the corresponding measured value.

Quantity	GTE	Measured	Agreement
Ω_Λ (PSC)	0.6899	0.6847 ± 0.0073	0.71σ
Ω_Λ (holographic)	0.6732	0.6847 ± 0.0073	-2.8σ
n_s	0.96488	0.9649 ± 0.0042	0.004σ
Σm_ν	59.4 meV	< 120 meV	factor 2 below
δ_{CP}	$68.51°$	$68.53°$	0.017%
J	3.02×10^{-5}	3.18×10^{-5}	5%
A	$\sin(\pi/3) = 0.866$	0.836 ± 0.015	within 2σ

Table 6: GTE predictions against current observation. Planck 2018 for Ω_Λ and n_s; PDG for the CKM quantities.

The ruled-out CMB mechanisms include the $\mathbb{Z}_7$ running route ($n_s(\mathbb{Z}_7) = -15.11\sigma$), which is excluded by the superselection argument of Section 4 in favor of the binary $\mathbb{Z}_2$ route.

Computational Tools

All numerical computations use Python 3.9+ with NumPy for linear algebra and array operations, and SciPy for integration and special functions. Key scripts used in this paper:

- `papers/45_three_tape_cmca/scripts/cc_temporal_voxel_formula.py` — temporal-voxel CC coefficient $\rho_\Lambda = (9/112)\, M_{\rm Pl}^2 H_0^2$ and $\Omega_\Lambda = 3\pi/14$.
- `papers/45_three_tape_cmca/scripts/gorard_planck_normalization.py` — Gorard coefficient $C_{\rm Gorard} = 3/32$ and Newton-constant normalization.
- `papers/18_koide_cyclotomic/scripts/gte_cp_phase_final.py` — CKM CP phase $\delta_{CP} = \pi/2 - 3/8$ and Jarlskog invariant J.
- `papers/21_neutrino_masses/scripts/gte_leptogenesis.py` — leptogenesis efficiency factor K_1 and baryon asymmetry feasibility.
- `papers/21_neutrino_masses/scripts/neutrino_mass_prediction.py` — neutrino mass spectrum and sum Σm_ν.

The machine-checked formalisation is written in Lean 4 with Mathlib. All theorems cited as $\mathbf{A}_{\rm Lean}$ carry zero `sorry` and zero custom axioms beyond Lean's standard foundations; named physics axioms are explicitly disclosed wherever used. The full theorem inventory is in Appendix A.

Full reproduction instructions are provided in `REPRODUCE.md` co-located with this paper's source.

AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations, and Lean 4 certifications were independently verified by deterministic scripts and the Lean 4 kernel.

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